

MECHANISED META-THEORY FOR A VERIFIED PROOF ASSISTANT

Meven LENNON-BERTRAND

CNRS Audition



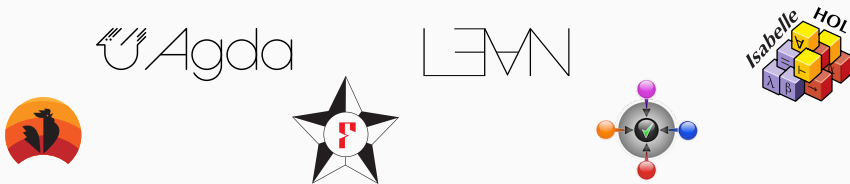


Agda

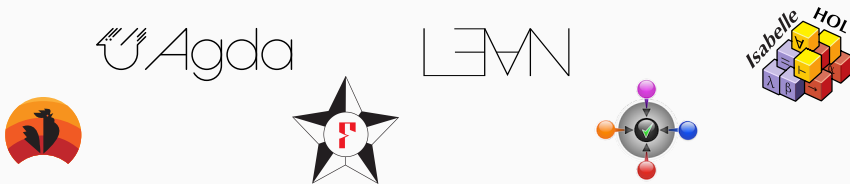


LEAN



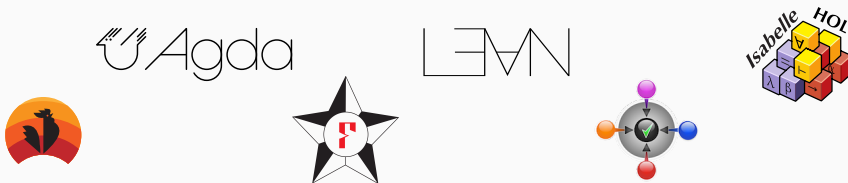


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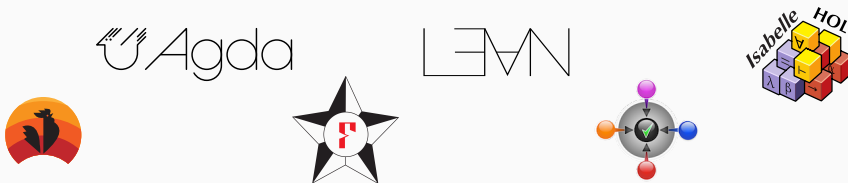
For informatics: mathematical certainty for software/hardware correctness, in academia and industry (Amazon, Chrome, WebAssembly, Airbus, Arm...)



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A rich and growing community of users, with diverse needs

Safety

vs

Expressivity



Safety

vs

Expressivity

Trust

Usability





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Feature Conservatism. Not all Coq features have the same degree of maturity. We favor stable, battle-tested features over experimental and deprecated ones.

ANSSI, Requirements on the Use of Coq in the Context of Common Criteria Evaluations

CS & Math masters

Agrégation (10th)

2019

PhD

2022

Post-doc



Gradual typing ——— TOPLAS ICFP ———→

Subtyping ——— ESOP⁹ →

ITP ——— META^{COQ} ——— WITS Tutorial @ POPL JACM¹ →

— LOGREL-COQ ——— CPP⁹ Preprint¹ →

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Concrete outputs & Ambitious theory

———— Gradual typing —————→

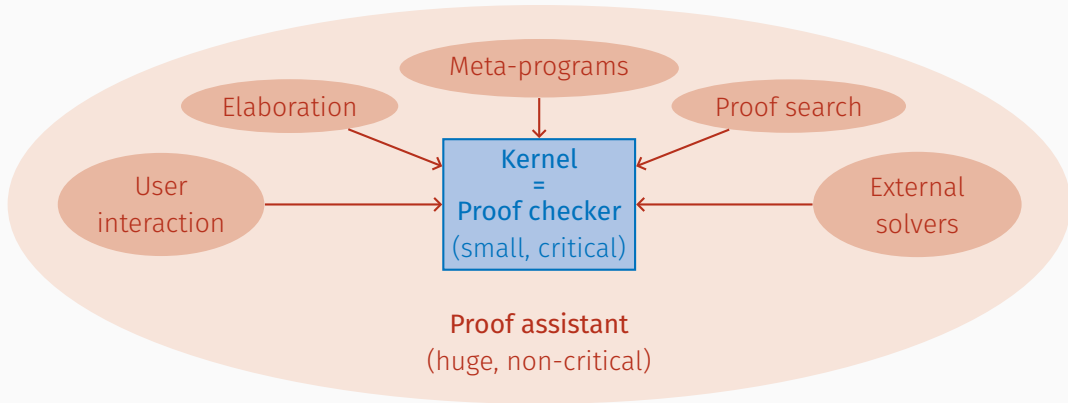
———— Subtyping —————→

———— Bidirectional typing —————→

———— META-CoQ —————→

———— LOGREL-CoQ —————→

ONLY TRUST THE KERNEL



Problem: small kernel = strong constraint

Problem: who checks the checker?

TOWARDS A VERIFIED KERNEL

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Paradigm change: ~~Trusted code base~~ → **program verification**

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~~Pen and paper proofs~~ based on **mechanised meta-theory**

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Larger, more expressive but
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Peut-on mécaniser un bon fragment de la logique d'un assistant à la démonstration dans un fragment à peine plus gros?

Xavier Leroy, Cours au Collège de France, 2020

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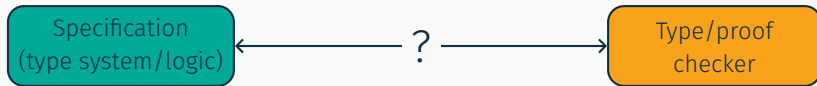
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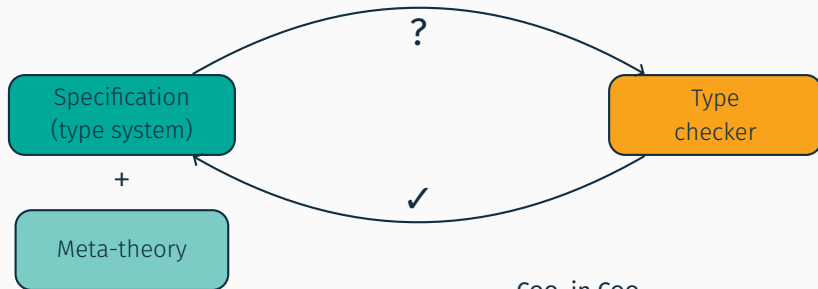
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Gödel incompleteness



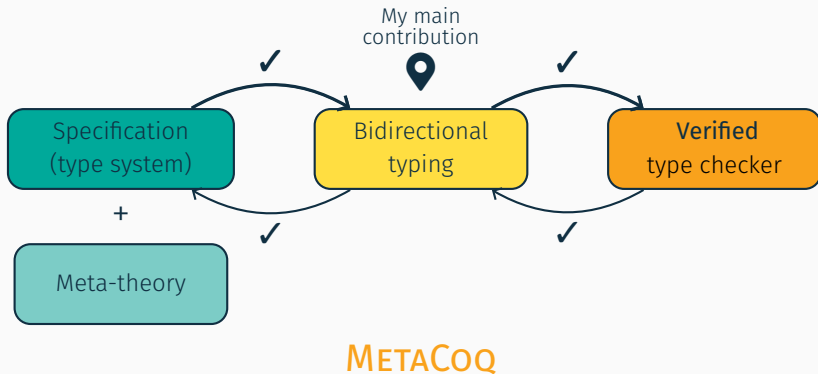


METACoQ

Coq, in Coq

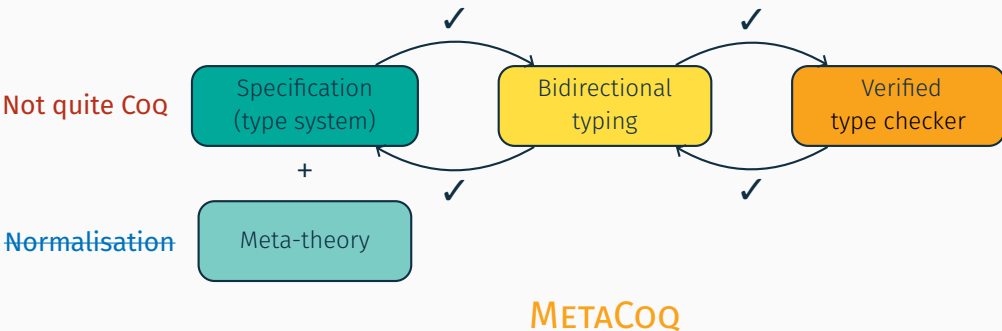
~10 core team members, >20 contributors

>200kLoC (~15kLoC by me)

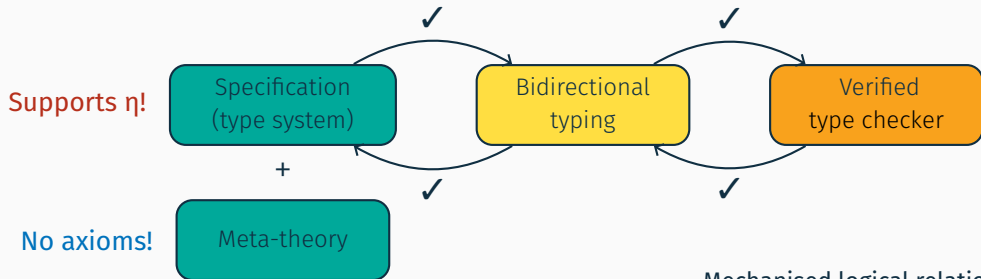


Journal of the ACM, tutorial @ POPL, Newton Institute...
Cross-fertilisation: impact on the CoQ kernel, LEAN4LEAN, AGDA-CORE...

METACoQ AND LOGREL-CoQ



But: **Missing features** because of **meta-theoretic** limitations
Axiomatised properties



LOGREL-CoQ

Mechanised logical relations

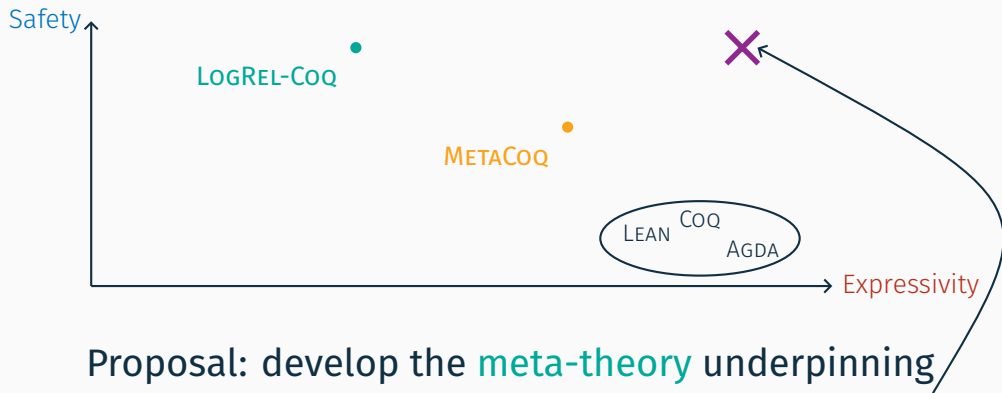
~3 active devs, I am the project lead
30kLoC (~20kLoC by me)

CPP[🐼], ESOP[🐼], LICS, preprint^{!🐼} ...

RESEARCH PROPOSAL

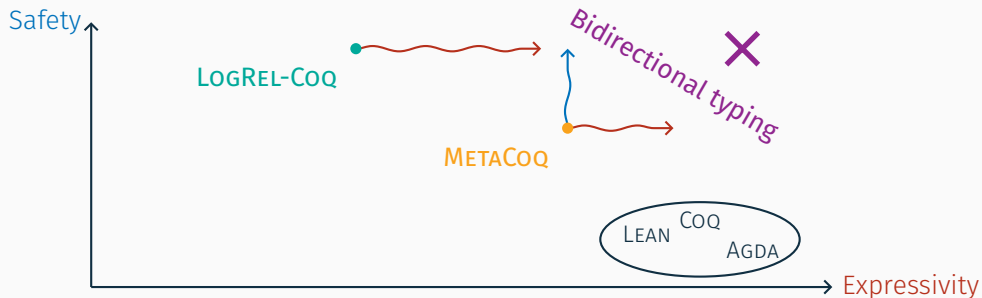


RESEARCH PROPOSAL



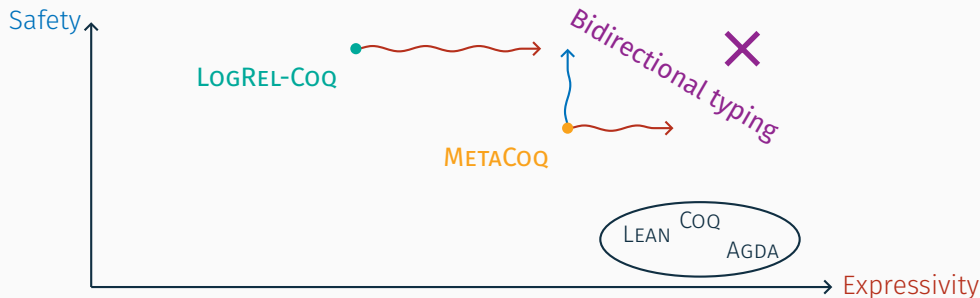
Proposal: develop the **meta-theory** underpinning
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Preuves & Programmes (IRIF), LoCal (LIPN), Plume (LIP), Preuves Mécanisées (LMF)

EXTENSIONALITY EQUATIONS

Short term [Herbelin, Melliès (IRIF), Winterhalter, Blanqui (LMF)]

$$f \cong \lambda x. f\ x$$

η (functions)

$$(\text{fst}(p), \text{snd}(p)) \cong p$$

η (pairs)

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A known difficulty in **meta-theory** (e.g. λ -calculus)...

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Approach: **typed** rewriting

TOWARDS A VERIFIED GUARD CONDITION

Medium term [Herbelin (IRIF), Pous, Zakowski (LIP), Paulin-Mohring (LMF)]

Normalisation: key property for **decidability**
logical consistency → incompleteness

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METACOQ

LEMN



```
fix f n :=  
match n with ...
```

+ guard condition

Agda

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METACoQ

LOGREL-CoQ

LEMN

normalisation
proofs



fix f n :=

match n **with** ...

+ guard condition

compilation

nat_rec

≈

prim. recursion

meta-theory

Agda

models

How to scale the **meta-theory** of LOGREL-COQ to **realistic languages**?

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Practical tooling

Short term [Laurent (LIP), Winterhalter (LMF)]

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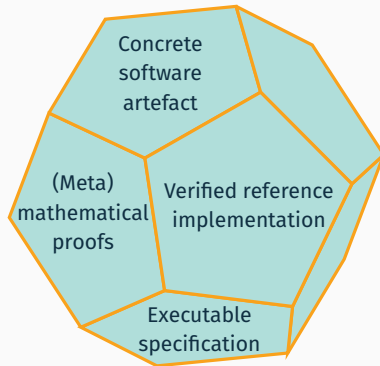
Theoretical foundations

Long term [Melliès, Scherer (IRIF),
Frey, Rogers (LIPN), Barras (LMF)]

Formalise the internal logic of
presheaf toposes?

IN CONCLUSION

A fundamentally **new, hybrid** object, between **theory** and **implementation**



with the potential to **deeply change** the proof assistant ecosystem.

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